The empirical recurrence for OEIS A260206, recovered from its tail

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Abstract

OEIS A260206 records “Empirical recurrence of order 80” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 124$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 84$. Its last coefficient is $c_{80} = 576 \neq 0$, so no further tail satisfies anything shorter; any order-80 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A260206 is “Number of (n+2)X(7+2) 0..1 arrays with each 3X3 subblock having clockwise perimeter pattern 00000000 00000001 or 00010001.”. It has offset 1 and begins

$$2274, 6895, 45181, 245486, 1473681, 8929505, 52387329, 310875809, \ldots$$

The entry states:

> Empirical recurrence of order 80 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 12:03:14, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 124$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 124$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 80: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 80.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 248$ exact terms from $n = 4$ on, it returns order 80 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{80} c_i\, a(n-i),$$

c_1	1	c_{21}	44436264	c_{41}	2236173507	c_{61}	−41920718
c_2	10	c_{22}	114687111	c_{42}	2554767458	c_{62}	−2556708
c_3	113	c_{23}	50107332	c_{43}	6890129224	c_{63}	−11231302
c_4	194	c_{24}	−83316716	c_{44}	−5957721496	c_{64}	−30970954
c_5	−364	c_{25}	−313598824	c_{45}	−193584220	c_{65}	7922349
c_6	−3274	c_{26}	−431680883	c_{46}	−7777795984	c_{66}	6799998
c_7	−9037	c_{27}	182203615	c_{47}	−3052744816	c_{67}	1554747
c_8	2945	c_{28}	1045280807	c_{48}	2463423975	c_{68}	6947158
c_9	59888	c_{29}	547320138	c_{49}	1358903677	c_{69}	−631484
c_{10}	155810	c_{30}	375210116	c_{50}	8217596583	c_{70}	−1148263
c_{11}	58987	c_{31}	−1850933876	c_{51}	2939515791	c_{71}	489817
c_{12}	−739332	c_{32}	−2328856527	c_{52}	3630304018	c_{72}	−520265
c_{13}	−2159980	c_{33}	818908824	c_{53}	863596131	c_{73}	72043
c_{14}	85052	c_{34}	359481517	c_{54}	−1192650468	c_{74}	156066
c_{15}	8630480	c_{35}	4650452202	c_{55}	−421639806	c_{75}	−27253
c_{16}	9966973	c_{36}	1501741438	c_{56}	−1408521627	c_{76}	12241
c_{17}	−11474578	c_{37}	−3097317999	c_{57}	−328822035	c_{77}	2368
c_{18}	−28026916	c_{38}	−674694755	c_{58}	−204176562	c_{78}	−5660
c_{19}	−15684798	c_{39}	−7361871814	c_{59}	−91182671	c_{79}	396
c_{20}	4322218	c_{40}	2345933441	c_{60}	133342681	c_{80}	576

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 84$, and it is the recurrence OEIS A260206 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{80} - c_1 x^{80-1} - \cdots - c_{80}$. Its constant term is $-c_{80} = -576$, which is not zero, so $q_{\min}(0) \neq 0$ and

$q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 80 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 80, so $\deg q = 80$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 80, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 18 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 80, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 576.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-80 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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